28/01/25 Monday.

CHEMICAL BONDING.

⇒ BOND ; Attraction b/w atoms or molecules

⇒ Types ; 1. Ionic
2. covalent
3. Metallic
4. coordinate.

⇒ Quantum no.s.

* n - principal QN - main energy levels - eg ; K, L, M, N

n = 1 - K shell $n \neq 0$

n = 2 - L shell

n = 3 - M shell

* l - Azimuthal QN - Subenergy level / subshell to which the ē belongs.

eg: S, P, d, f, ..

l ; $0 \rightarrow n-1$

n = 1 , l = 0 → S

n = 2 , l = 0, 1 → S, P

n = 3 , l = 0, 1, 2 ⇒ S, P, d

n = 4 , l = 0, 1, 2, 3 ⇒ S, P, d, f

* m - Magnetic QN - no. of orientation of orbitals in a subshell.

□ S l = 0 , m = 0 (Spherically symmetrical (only one orientation))

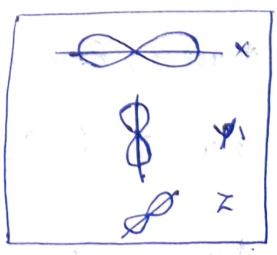
m ; -l to +l incl : 0

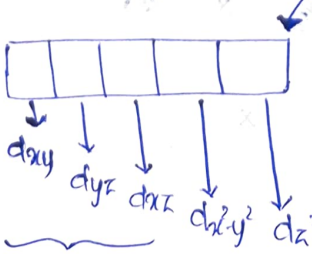
P l = 1 , m = -1, 0, 1

d l = 2 , m = -2, -1, 0, 1, 2

f l = 3 , m = -3, -2, -1, 0, 1, 2, 3

no. of total values ; 2l + 1

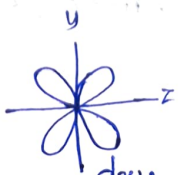


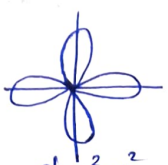


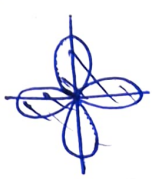


in b/w axes

along the axes







→ All the d-orbitals have same energy ; ∵ they are degenerate.

* Spin QN → s → spinning of each electron within orbital.

+½ - clockwise

-½ - anti-clockwise.

↑↓ net Spin = 0

29/07/25 Tuesday:

Q: Identify the correct set of quantum no.s from the following?

* $3P_x^1$ $3P_y^1$ $3P_z^1$

$n=3$	$n=3$	$n=3$
$l=1$	$l=1$	$l=1$
$m=+1$ or 0 or -1	$m=0$	$m=1$
$s=+1/2$	$s=+1/2$	$s=+1/2$

1	1	1
$3P_x$	$3P_y$	$3P_z$

* $3P_x^2$ $3P_y^1$ $3P_z^1$

$n=3$	$n=3$	$n=3$
$l=1$	$l=1$	$l=1$
$m=-1$	$m=0$	$m=+1$
$s=-1/2$	$s=+1/2$	$s=+1/2$

2nd ē

Q Identify the correct set of QNs:

* $n=3$, $l=2$, $m=0$, $s=+1/2$ ✓

* $n=4$, $l=0$, $m=-1$, $s=-1/2$ ×

* $n=3$, $l=2$, $m=-2$, $s=-1/2$ ✓

* $n=5$, $l=0$, $m=-1$, $s=+1/2$ ×

* $n=6$, $l=4$, $m=0$, $s=0$ ×

Q: Write the possible quantum no. values of;

1. $4s^2$
2. $3P_z^1$
3. $2p^3$

1.

1st ē	2nd ē
$n=4$	$n=4$
$l=0$	$l=0$
$m=0$	$m=0$
$s=+1/2$	$s=-1/2$

2.

$n=3$
$l=1$
$m=+1$
$s=+1/2$

3.

1st ē	2nd ē	3rd ē
$n=2$	$n=2$	$n=2$
$l=1$	$l=1$	$l=1$
$m=-1$	$m=0$	$m=+1$
$s=+1/2$	$s=+1/2$	$s=+1/2$

⇒ Aufbau Principle.

- ēs are filled in the orbital in the increasing order of energy according to their '$(n+l)$' value.
- if $(n+l)$ value same for some reason; the one with lowest 'n' will be filled first.

→ Pauli's Exclusion Principle.

- no two ēs in an atom can have same set of all four quantum no.s.

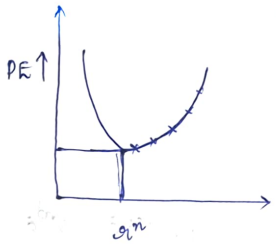
⇒ Hund's Rule Of Maximum Multiplicity.

- pairing will take place only when all the orbitals are singly occupied.

30/07/2025 Wednesday.

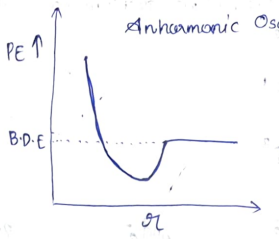
⇒ Energetics Of Chemical Bonding:

When two Hydrogen atoms are coming closer to form H_2 through covalent bond.



Theoretical diagram (graph)

Anharmonic Oscillators.



Practical graph.

* For harmonic oscillators (Theoretical) the molecules can be compressed and stretched to maximum in their bond.
* But in reality, anharmonic oscillators donot stretch that much and will break bonds at Bond Dissociation Energy.

⇒ Ionic Bonding:

* The formation of an ionic bond is accompanied by decrease of energy.
* One atom completely loss $\bar{e}$ and other gain an $\bar{e}$
  - loss → cation
  - gain → anion
* The strong electrostatical interaction which exists b/w constituent ions forms ionic bond.

⇒ Energetics of Ionic Bond formation:

eg.: Na^+Cl^-

$$Na_{(s)} \xrightarrow{\Delta_s H} Na_{(g)} \xrightarrow{IE} Na^+_{(g)}$$

* Ionisation energy:
  required to remove the most loosely bonded $\bar{e}$ from the gaseous valence shell of a gaseous isolated atom

$$\tfrac{1}{2}Cl_{2(g)} \xrightarrow{BDE\ (+ve)} Cl_{(g)} \xrightarrow{EA\ (-ve)} Cl^-_{(g)}$$

* Enthalpy of dissociation

Amnt of energy required to break 1 mole of a bond.

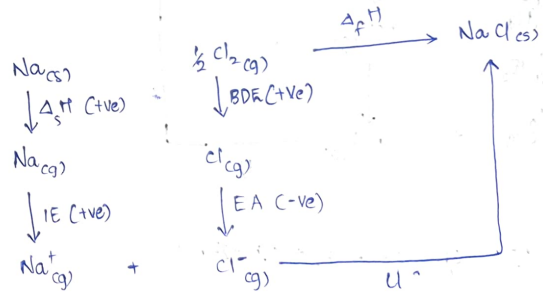
* Electron affinity

Amnt of energy released when an extra $\bar{e}$ is added to the valence sheel of an isolated gaseous atom.

$$Na^+_{(g)} + Cl^-_{(g)} \xrightarrow{U\ (-ve)} NaCl_{(s)}$$

* Lattice Energy (U) (Always -ve)

Amnt of energy released when ca 1 mole of a solid ionic compd is formed from its gaseous ions.



* Enthalpy of formation ($\Delta_f H$)

The enthalpy change when one mole of a compound is formed from its elements in their standard states.

→ Born-Haber Cycle.

* Used to find Lattice Enthalpy, indirectly using Hess's Law.

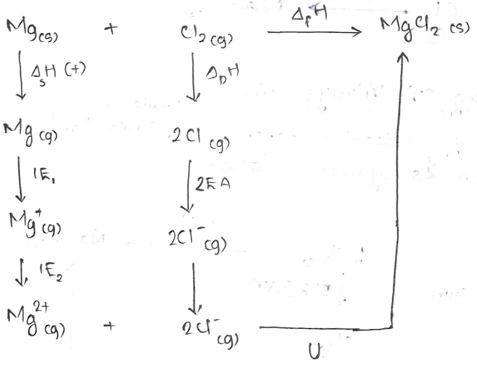
→ Hess's Law.

* The enthalpy of rⁿ is same if the chemical rⁿ is taking place in single step or multiple steps.

$$\Delta_f H = \Delta_s H + IE + \tfrac{1}{2}\Delta_D H + EA + U$$

$$U = \Delta_f H - [\Delta_s H + IE + \tfrac{1}{2}\Delta_D H + EA]$$

Q:- Born-Haber cycle for $MgCl_2$.



$$\Delta_f H = \Delta_s H + IE_1 + IE_2 + \Delta_D H + 2EA + U$$

Q:- Calculate the lattice energy of Magnesium fluoride (MgF_2)

ΔH_s = 146.4 kJ/mol

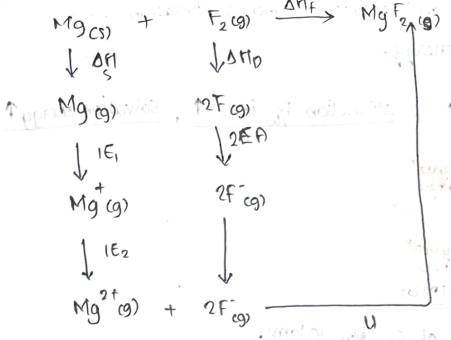
$\Delta H_D (F_2)$ = 158.8 kJ/mol

IE_1 (Mg) = 737 kJ/mol

IE_2 (Mg) = 1449 kJ/mol

$EA (F)$ = −328 kJ/mol

$\Delta H_f (MgF_2)$ = −1096.5 kJ/mol



$$\Delta H_{lat} = U = \Delta H_f - [\Delta H_s + IE_1 + IE_2 + \Delta_D H + 2EA]$$

$$= -2931.7 \text{ kJ/mol}$$

⇒ Lattice Enthalpy

The energy provided / required (change) when 1 mole of an ionic solid is formed from its gaseous ions.

★ Lattice enthalpy is equal and opposite to lattice energy.

* Solubility of any compound in water depends on two factors;

1. Solvation energy - high.
2. Lattice energy - low

Solvation Energy & Solubility of Ionic Solids

* Ionic compounds are only very slightly soluble in most common solvents, except for those that are quite polar.

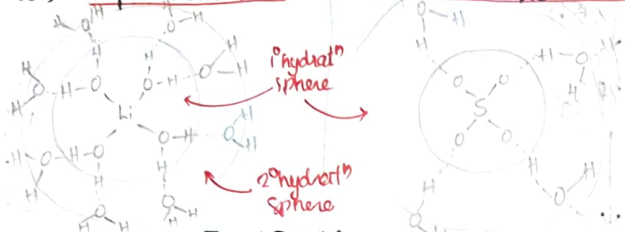
* Solution occurs if attractive force b/w ions and solvent molecules are comparable to lattice energy.

* The energy released during solvation of ions is called solvating ener

* an ionic crystal will dissolve when ; solvation energy overcomes lattice energy.

* as dipolemoment ↑ , attraction b/w ions ↑ , solvation energy ↑

*



4/8/25 Monday

⇒ COVALENT BONDING:

* Formed by the sharing of $\bar{e}$s b/w atoms. (mutual)

⇒ Valence Bond Theory (V.B.T)

* proposed by Linus Pauling.
* bond formed by Overlapping of half filled atomic orbitals of combining atoms.
* The partial interpenetration of atomic orbitals is known as orbital Overlap.

eg:- H H

1S' 1S'

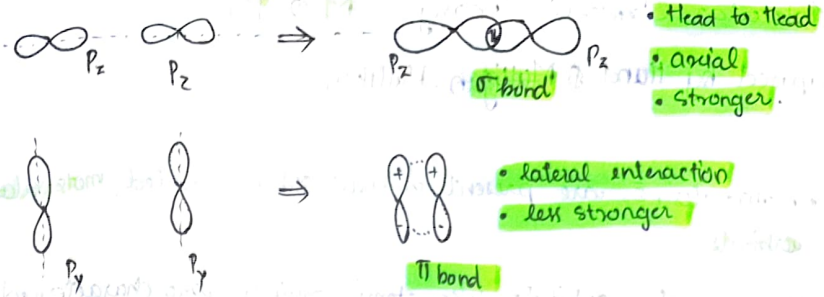


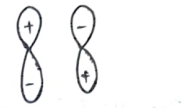
H-H

* S subshell ⇒ 
* p subshell ⇒ 

* overlapping should be done with same charges



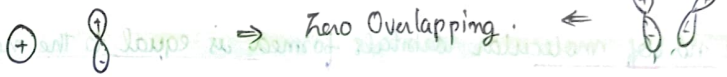




⇒ negative overlapping

⇓

no bond formation.



⇒ Zero Overlapping. ⇐

Head on Head

H_2 

F_2 

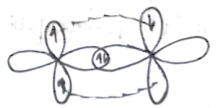
HF 

O_2



O O

⇓

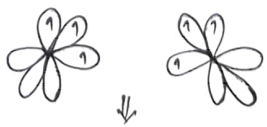


O=O

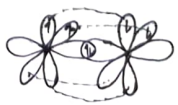
N_2 $2s^2\ 2p^3$



N N



⇓



N≡N

⇒ Molecular Orbital Theory (M.O.T)

* proposed by Hund & ~~Mulliken~~ Mulliken.
*
- In molecules, ēs are present in new orbitals called molecular orbitals.
- The molecular orbitals, like atomic orbitals, are characterised by a set of quantum no.s. They are formed by the combination of atomic orbitals of nearly same energy.
- The molecular orbitals are associated with the nuclei of all the atoms constituting the molecule.
- The no. of molecular orbitals formed is equal to the no. of atomic orbitals combined.
- The molecular orbitals ~~formed~~ are filled in the increasing order of their energies (Aufbau principle).
- A molecular orbital, like atomic orbitals, can accomodate only two ēs and these ēs must have opposite spins (Pauli's principle)
- refer Hund's rule;

LCAO (Linear Combination of atomic Orbitals)

* when two ~~orbitals~~ atomic orbitals combine to form molecular orbital, they can do so in two ways
  i) addition.
  ii) Subtraction.

i) By addition; $\Psi_{MO} = \Psi_A + \Psi_B$

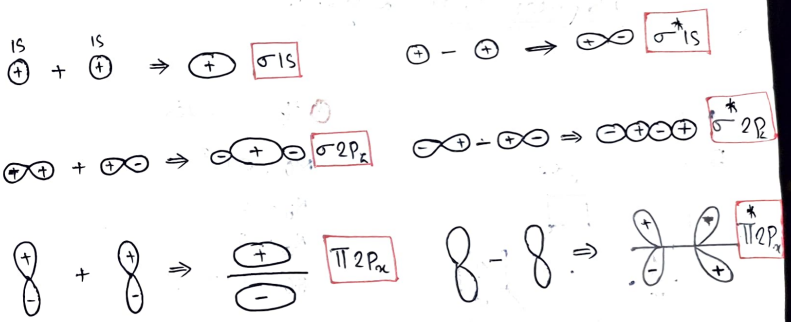
* lower energy M.O formed.
* Favours bonding ⇒ Bonding Molecular Orbital.

Ψ → orbital wave fn
probability of finding an ē in a specific region around nucleus of atom.

ii) By Subtraction;

$$\Psi^*_{MO} = \Psi_A - \Psi_B$$

* Higher energy than combining orbitals.
* Opposes bond formation ⇒ Antibonding Molecular Orbital.



5/08/25
Tuesday.

Molecular Orbital Configuration.

more than 14 ē system

$\sigma 1s \;\; \sigma^*1s \;\; \sigma 2s \;\; \sigma^*2s \;\; \sigma 2p_z \;\; \pi 2p_x = \pi 2p_y \;\; \pi^* 2p_x = \pi^* 2p_y \;\; \sigma^* 2p_z$

upto 14 ē system:

$\sigma 1s \;\; \sigma^*1s \;\; \sigma 2s \;\; \sigma^*2s \;\; \pi 2p_x = \pi 2p_y \;\; \sigma 2p_z \;\; \pi^* 2p_y = \pi^* 2p_y \;\; \sigma^* 2p_z$

reason; The interaction b/w these orbitals is more in less ē species
∴ $\sigma 2p_z$ energy is raised.

⇒ Bond Order.

* The no. of bonds present in b/w atoms in the molecule.
* gives the strength and stability of the molecule.

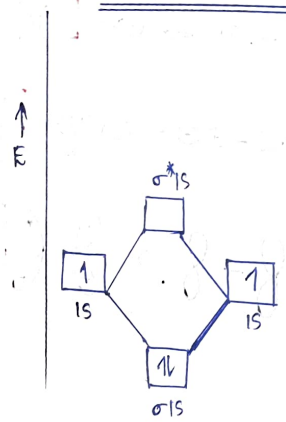
$$B.O = \frac{1}{2}\left[N_b - N_a\right]$$

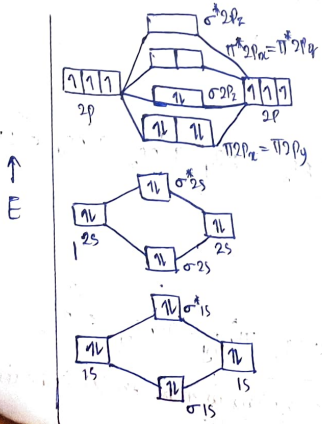
N_b ⇒ no. of $\bar{e}$s in B.M.O

N_a ⇒ no. of $\bar{e}$s in A.B.M.O (Starred)

* B.O is directly proportional to Bond dissociation energy.
* $B.O \propto \frac{1}{B.L}$ Bond length inversely proportional.
* B.O ∝ Stability.

Molecular Orbital Diagram.





⊛ H_2 ≡ 2 $\bar{e}$s

MO : $\sigma 1s^2$

$N_b = 2$

$N_a = 0$

$B.O = \frac{1}{2} \times 2 = 1$

H–H

⊛ N_2 14 $\bar{e}$ ≡ $1s^2\ 2s^2\ 2p^3$ each

MO : $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2$

$\pi 2p_x^2 = \pi 2p_y^2\ \sigma 2p_z^2$

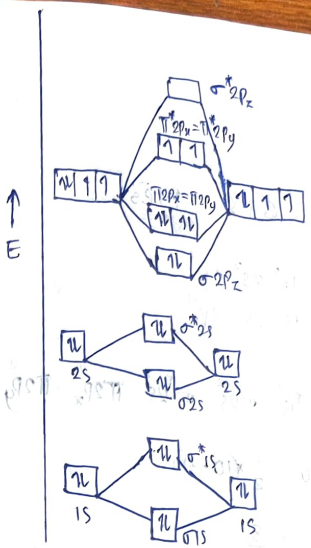
$N_b = 10$

$N_a = 4$

$B.O = \frac{1}{2}(10-4)$

$= 3$

N≡N



06/08/25
Wednesday.

⊛ O_2 ⇒ $1s^2\ 2s^2\ 2p^4$ each = 16 $\bar{e}$s

MO: $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2 = \pi$

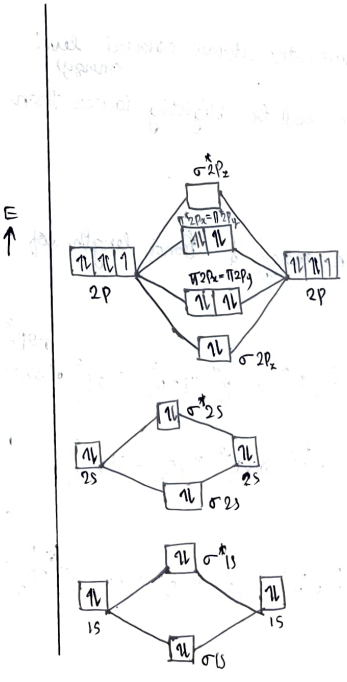
$\pi^* 2p_x^1 = \pi^* 2p_y^1$

$N_b = 10$

$N_a = 6$

$B.O = \frac{1}{2}(10-6) = \frac{1}{2} \times 4 = 2$

O=O



⊛ F_2 ⇒ $1s^2\ 2s^2\ 2p^5$ ⇒ 18 $\bar{e}$s each

M.O : $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2$

$\pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_x^2 = \pi^* 2p_y^2$

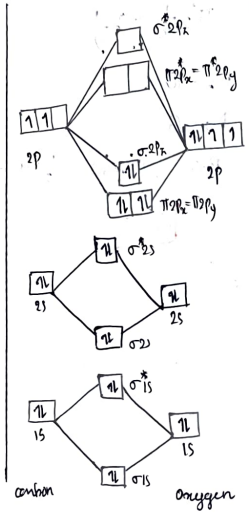
$N_b = 10$

$N_a = 8$

$B.O = \frac{1}{2} \times 2 = 1$

F–F

⇒ HETERONUCLEAR MOLECULE



* CO

no. of $\bar{e}$s = 6 + 8 = 14 $\bar{e}$s

C ⇒ $1s^2\ 2s^2\ 2p^2$

O ⇒ $1s^2\ 2s^2\ 2p^4$

$\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2\ \sigma 2p_z^2$

B.O = $\frac{1}{2} \times (10 - 4)$

= 3

C ≡ O

"when we have two diff atoms, the atomic orbital level (energy) of more electronegative atom will be slightly lower than the other."

⇒ Compare the Bond dissociation energy & Bond length of N_2, N_2^+, N_2^-, N_2^{2+}, N_2^{2-}

ans:- N_2 - 14$\bar{e}$ - $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2\ \sigma 2p_z^2$

B.O = 3

N_2^+ - 13$\bar{e}$ - $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2\ \sigma 2p_z^1$

B.O = $\frac{1}{2} * 5 = 2.5$

N_2^- - 15$\bar{e}$ - $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_x^2\ \pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_x^1 = \pi^* 2p_y$

B.O = 2.5

N_2^{2+} ⇒ 12$\bar{e}$ $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \pi 2p_x^2 = \pi 2p_y^2$

B.O = 2

N_2^{2-} ⇒ 16$\bar{e}$ $\sigma 1s^2\ \sigma^* 1s^2\ \sigma 2s^2\ \sigma^* 2s^2\ \sigma 2p_z^2\ \pi 2p_x^2 = \pi 2p_y^2\ \pi^* 2p_x^1 = \pi^* 2p_y^1$

B.O = 2

B.D.E ; $N_2 > N_2^+ > N_2^- > N_2^{2+} > N_2^{2-}$

B.L ; $N_2^{2-} > N_2^{2+} > N_2^- > N_2^+ > N_2$

B.D.E $\propto \frac{1}{B.L}$

08-08-2025
Friday. ⇒ Fajan's Rule.

* explains the covalent partial covalent character of Ionic bond.
* depends upon ;
  1. polarising power of cation.
  2. Polarisability of anion.

* down the grp polarising power of cation decreases.

Li^+
Na^+
K^+ ↓ polarising power ↓
Rb^+
Cs^+

* down the grp, polarisability of anion increases.

F^-
Cl^- ↓ polarisability ↑
B^-
I^-

Covalent nature:

- $Li^+ < Be^{2+} < Al^{3+}$
- $LiCl < AlCl_3$
- $NaCl > RbCl$
- $BeCl_2 > MgCl_2$
- $NaCl < MgCl_2$

- Smaller the size of cation, larger the charge of cation;
- Larger the size of anion, larger the charge of anion;

Covalent character of compound ↑

⇒ Partial Ionic character of Covalent bond / polarity of Covalent bond.

* arises due to the difference in electronegativity b/w atoms
* polarity is measured by dipolemoment (μ)

$\mu = q \times d$

q: charge; d: distance b/w atoms

* $\overset{\delta^+}{H}-\overset{\delta^-}{Cl}$
* % ionic character $= \frac{\mu_{obs}}{\mu_{ionic}} \times 100$

* if $\mu = 0$, the molecule will be symmetric.
* even if CO_2 bonds have dipolemoment, since they cancel each other, it doesnot have net dipolemoment & hence is symmetric.

$O \overset{\longleftarrow}{=} C \overset{\longrightarrow}{=} O \quad \mu_{net} = 0$

Q: The inter nuclear distance b/w H & Cl is 129 pm. The actual dipolemoment is 3.44×10^{-30} Cm. Calculate the % ionic character.

Coloumbmeter.

% ionic character $= \frac{\mu_{obs}}{\mu_{ionic}} \times 100$

$= \frac{3.44 \times 10^{-30}}{q \times 127 \times 10^{-12}} \times 100$

$= \frac{3.44 \times 10^{-30}}{1.6 \times 10^{-19} \times 127 \times 10^{-12}} \times 100 = \frac{3.44 \times 10^{-30}}{2.032 \times 10^{-40}} \times 100$

$= 0.169 \times 100$

$= 16.9$

$= 17\%$

⇒ HYBRIDISATION:

intermixing of atomic orbitals of slightly different energy so as to redistribute their energies and to give new set of orbitals of equivalent energies and shape.

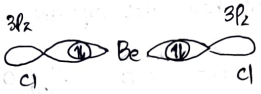
eg: sp, sp^3d, sp^2, sp^3d^2, sp^3, sp^3d^3

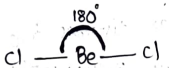
sp — $BeCl_2$

Be $2s^2\ 2p^0$

ground state: 2s | 2p

excited state: 2s | 2p → sp





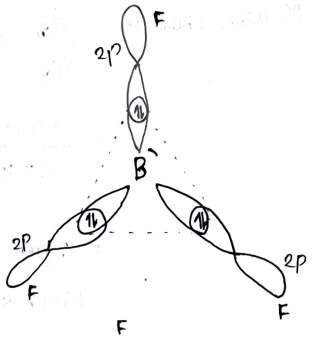
linear

sp^2 — BF_3

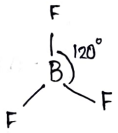
B - $2s^2\ 2p^1$

G.S - 2s | 2p

E.S - 2s | sp^2 | 2p



trigonal planar

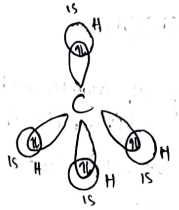


sp^3 — CH_4

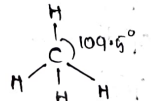
C - $2s^2\ 2p^2$

G.S - 2s | 2p

E.S - sp^3



tetrahedral

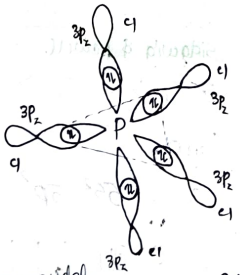


sp^3d — PCl_5

P - $3s^2\ 3p^3\ 3d^0$

G.S - 3s | 3p | 3d

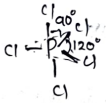
E.S - sp^3d



Trigonal bipyramidal

axial = 90°

equi = 120°

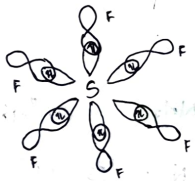


sp^3d^2 — SF_6

S - $3s^2\ 3p^4\ 3d^0$

G.S - 3s | 3p | 3d

E.S - sp^3d^2



octahedral

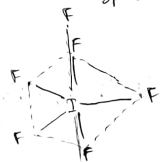


sp^3d^3 — IF_7

I - $5s^2\ 5p^5\ 5d^0$

G.S - | | | |

E.S - sp^3d^3



pentagonal bipyramidal

axial ⇒ 90°

interhalogen compounds

ICl_5 ; IF_7

BrF_3 $BrCl_3$

ClF_3 ...

Valence Shell Electron Pair Repulsion Theory (VSEPR)

Nyholm Tuckett ⇒ Sidgwig & powell ⇒ Based „on repulsive interactions of e pairs in valence shell of atoms.

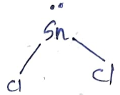
* lp-lp > lp-Bp > Bp-Bp.

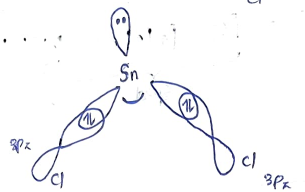
* $SnCl_2$

Sn - $5s^2\ 5p^2$

G.S - [↑↓] [↑] [↑]

sp^2





Bond angle is less that 120° due to lp-Bp repulsion.

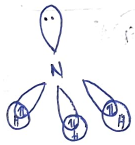
Bent shape. / V-shape.

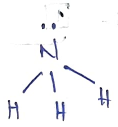
* NH_3 107°48'

N - $2s^2\ 2p^3$

G.S - [↑↓] [↑] [↑] [↑]

sp^3





inverted umbrella.

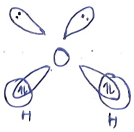
Pyramidal

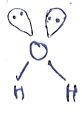
* H_2O

O - $2s^2\ 2p^4$

G.S - [↑↓] [↑↓] [↑] [↑]

sp^3





V-shape / Bent shape.

• Bond angle approx 104°. Pairs of lp-lp repulsion, Bp's comes closer.

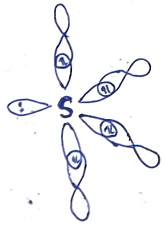
* SF_4

S - $3s^2\ 3p^4$

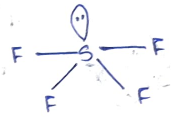
G.S - [↑↓] [↑↓] [↑] [↑] [ ] [ ] [ ] [ ]

E.S - [↑↓] [↑] [↑] [↑] [↑] [ ] [ ] [ ] [ ]

sp^3d



See saw shape



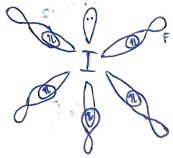
* IF_5

I - $5s^2\ 5p^5$

G.S - [↑↓] [↑↓] [↑] [↑] [↑]

E.S - [↑↓] [↑] [↑] [↑] [↑] [↑↓] [ ] [ ] [ ]

sp^3d^2





square pyramidal.

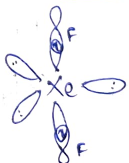
* XeF_2

Xe - $5s^2\ 5p^6$

G.S - [↑↓] [↑↓] [↑↓] [↑↓] [ ] [ ] [ ] [ ] [ ]

E.S - [↑↓] [↑↓] [↑↓] [↑] [↑] [ ] [ ] [ ] [ ]

sp^3d

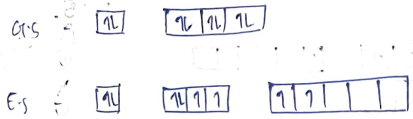




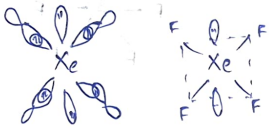
Linear geometry.

* XeF_4

Xe = $5s^2$ $5p^6$



sp^3d^2



Square planar.

13/08/25
Wednesday.

Bond pair	Lone pair	Hybridisation	Geometry	Example.
2	0	sp	linear	$BeCl_2$
2	1	sp^2	Bent	$SnCl_2$
2	2	sp^3	Bent	H_2O
2	3	sp^3d	linear	XeF_2
3	0	sp^2	Trigonal planar	BF_3
3	1	sp^3	Pyramidal	NH_3
3	2	sp^3d	T-shape.	ClF_3
4	0	sp^3	tetrahedral	CH_4
4	1	sp^3d	See-saw	SF_4
4	2	sp^3d^2	Square planar	XeF_4.

13/08/25
Monday.

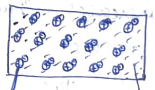
Metallic Bond.

* properties of metals :
  1. good conductors of heat & electricity.
  2. lustre
  3. malleability.
  4. Ductility.
  5. High melting and boiling point.
  6. Very low ionisation energy.

* Various theories have been proposed to account for the nature of bonding in metals.

1 The Classical Free-Electron Theory

* proposed by Drude and Lorentz.
* idea based : high electrical and thermal conductivities of metals are due to the presence of free $\bar{e}$s.
* A metal behaves as if it is an assemblage of +ve ions (core or kernel) immersed in a sea of mobile $\bar{e}$s.
* A force of attraction which binds a metal ion (kernel) to a number of $\bar{e}$s within its sphere of influence is known as metallic bond.



metallic kernel. sea of $\bar{e}$s

* This theory explains :
  - Electrical and thermal conductivity. (due to presence of free $\bar{e}$s, move)
  - metallic lustre. : due to mobile free $\bar{e}$s. when light fall on metal surface, $\bar{e}$s move to & fro ∴ moving charge emits (produces) EMR in form of light. It just appears to be reflected.

- malleability & ductility : Since metallic bond is non-directional
- High tensile strength : can resist stretching without breaking due to strong electrostatic force of attraction.
- Elasticity : movement of metal ions.

* limitation – couldn't explain relative difference in certain elemental properties.

II The Valence Bond Theory.

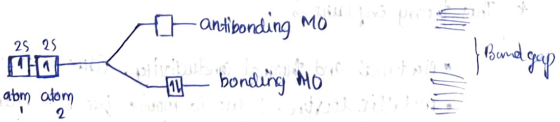
* developed by pauling.
* postulates that

'Resonance permits the existence of normal covalencies which ~~resonate among the alternate interatomic positions in the metal~~ be resonance between all the possible equivalent pairs of atoms, even if the number of nearest neighbours of an atom in a metal exceeds its valence es.'

III The Band Theory - Molecular Orbital Theory Of Metals.

* The e·c of lithium is $1s^2\ 2s^1$. when two lithium atoms approach each other, the 2s orbital of one atom can overlap with 2s orbital of another atom to form two molecular orbitals, one with lower and one with higher energy than combining orbitals.
* For n atoms, n molecular orbitals will be formed.
* as the no·of molecular orbitals increases, spacing b/w energy levels ~~decreases~~ of various orbitals decreases.
* when large no·of atoms, energy levels close together, they almost forms an 'energy band' or 'energy continuum'.

eg:-

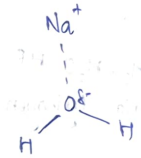


⇒ Secondary Bond forces.

* The intermolecular forces by which molecules are held together is called secondary bond forces.
* intermolecular forces are known as van der Waals' forces.

1. Ion dipole interactions.

- responsible for the solvation of ions by polar solvents and dissolution of ionic substances in such solvents.



2. Dipole - Dipole interaction

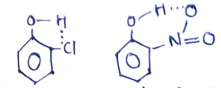
- The attractive forces due to the electrical interaction between the dipoles is known as dipole - dipole interaction.
- eg: NH_3 (ammonia), SO_2, HCl, HF, etc have permanent dipoles · so have d-d interaction.

- H bond

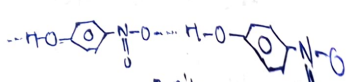
The attractive force which binds hydrogen atom of one molecule with electronegative atom of same molecule or another molecule of same substance or another substance.

---H-X---H-X----H-X---H-X---

intramolecular



intermolecular



* intramolecular H bonding is commonly

* As a result of intermolecular H bonding, the para derivatives undergo association, resulting in an increase in molecular mass and hence an increase in boiling point.

* Ortho derivatives are more volatile than para derivatives.

{ortho nitrophenol - volatile in steam
p-nitrophenol - non volatile in steam } ∴ steam distillation

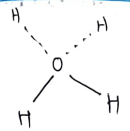
⇒ Consequences of H bonding.

- The boiling points of NH_3, H_2O, HF are abnormally high due to strong intermolecular hydrogen bonding.

18/08/25
Tuesday

? Why ice less denser than water.

In the solid phase ice forms a tetrahedral cage like structure which has more volume.



? ice when heated from 0°C ⟶ 4°C . contracts.

in this temp the cage str. H bonds breaks and therefore the volume decreases ie compression.

From 4°C onwards vapour pressure ↑, ... volume ↑.

3. Ion induced dipole interaction.

eg: Triiodide I_3^-

I^- + $I-I$ ⟶ I_3^-
iodide, iodine

eg: the approach of iodide ion, their is an induction of polarity in non-polar iodine (I_2) molecule ⇒ Electromeric effect

$I^- \rightarrow I-I \longrightarrow \left[I^{\delta-} \cdots I^{\delta+}-I^{\delta-}\right] \Rightarrow I_3^-$

4. dipole induced dipole interaction.

- eg:- Hydrated compound. {dipole inducing polarity in a non-polar molecule}
  Simple association to the molecule. no bonds or anything.
- hydrated Xenon compounds.

5. induced dipole - induced dipole interaction.

- London force or Van der waals force.
- simple interaction. Very weak.
- eg: $I^{\delta+}-I^{\delta-} \cdots I^{\delta+}-I^{\delta-}$

  due to the continous oscillation of $\bar{e}$s b/w the atoms, it sometimes comes closer to any one of the atom and hence induced dipole formed. and it induced the dipole polarity in next molecule.